

WORK SHEET - 05

STRAIGHT LINE AND CIRCLE

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.9 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 The least positive integral value of 'b' for which the point $(2b + 3, b^2)$ lies above the line $3x - 4y - a(a - 2) = 0 \forall a \in \mathbb{R}$, is
(A) 1 (B) 2 (C) 3 (D) 4
- Q.2 The line $3x - 4y + 7 = 0$ is rotated through an angle of $\frac{\pi}{4}$ in the clockwise direction about $P(-1, 1)$. The equation of line in its new position is
(A) $x + 7y - 6 = 0$ (B) $5y - x - 6 = 0$ (C) $7y + 13x + 6 = 0$ (D) $x + y = 0$
- Q.3 The equations of perpendicular bisectors of two sides AB and AC of a triangle ABC are $x + y + 1 = 0$ and $x - y + 1 = 0$ respectively. If circumradius of $\triangle ABC$ is 2 units and the locus of vertex A is $x^2 + y^2 + gx + c = 0$, then $(g^2 + c^2)$, is equal to
(A) 4 (B) 5 (C) 9 (D) 13
- Q.4 If the circles $x^2 + y^2 + 2ax + 2by + c = 0$ and $x^2 + y^2 + 2bx + 2ay + c = 0$ where $c > 0$, have exactly one point in common then the value of $\frac{(a+b)^2}{2c}$ is
(A) 1 (B) $\sqrt{2}$ (C) 2 (D) $1/2$
- Q.5 The locus of the centres of circles passing through the origin and intersecting the circle $x^2 + y^2 - 5x + 3y = 1$ orthogonally, is
(A) circle of radius 1 (B) circle of radius $1/2$
(C) straight line of slope $3/5$ (D) straight line of slope $5/3$
- Q.6 If circular arcs $\widehat{AC}$ and $\widehat{BC}$ have centres at $B(-\alpha, 0)$ and $A(\alpha, 0)$ respectively and equation of circle which touches both arcs $\widehat{AC}$ and $\widehat{BC}$ and line AB is $(x - a)^2 + (y - b)^2 = r^2$, $r > 0$, if length of arc $BC = 8\pi$, the value of $|a + b + r - \alpha|$ equals
(A) 2 (B) 3 (C) 5 (D) 6
- Q.7 The number of integral value(s) of p for which the point $([2p - 4], \{3p^2 - 2p + 1\})$ lies inside the circle $x^2 + y^2 = 9$ is(are)
[Note : $[y]$ and $\{y\}$ denote greatest integer and fractional part functions of y respectively.]
(A) 2 (B) 3 (C) 4 (D) 5
- Q.8 The members of a family of circles are given by the equation $2(x^2 + y^2) + \lambda x - (1 + \lambda^2)y = 10$. The number of circles belonging to the family that are cut orthogonally by the fixed circle $x^2 + y^2 + 4x + 6y + 3 = 0$, is
(A) 2 (B) 1 (C) 0 (D) 3
- Q.9 Given 2 points A $(-2, 0)$ and B $(0, 4)$, then sum of abscissa and ordinate of the point C on the line $x - y = 0$ so that perimeter of $\triangle ABC$ is least is
(A) -2 (B) 0 (C) 1 (D) 2

[PARAGRAPH TYPE]

Q.10 to Q.14 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 10 & 11

A straight line $L : 4x - 4y + 3 = 0$ is rotated in clockwise about the point where the line cuts the y-axis and a circle S_1 whose centre is $\left(\lambda, \frac{3}{4}\right)$ touches both the lines L and L_1 (L_1 is the line obtained after rotation) and the x-axis.

- Q.10 The value of $[\lambda]$ is equal to
 (A) 1 (B) 2 (C) 8 (D) 9
 [Note : $[k]$ denotes greatest integral value of k .]
- Q.11 If area of the triangle formed by the lines L_1 , angle bisector between L & L_1 and the x -axis is $\frac{p}{q}$, $p, q \in \mathbb{N}$ then least value of $(p + q)$ equals
 (A) 3 (B) 17 (C) 41 (D) 56

Paragraph for question nos. 12 to 14

A straight line L with negative slope passes through the point $(9, 4)$ and cuts the positive coordinate axes at points P and Q respectively.

- Q.12 The minimum value of $OP + OQ$, as L varies, where O is the origin is
 (A) 18 (B) 25 (C) 36 (D) 49
- Q.13 The area of triangle OPQ , when $OP + OQ$ becomes minimum (where O is the origin) is
 (A) 75 (B) 90 (C) 125 (D) 150
- Q.14 Let R be a moving point on xy plane such that $OPRQ$ becomes a rectangle then the locus of R , as L varies is
 (A) $\frac{x}{9} + \frac{y}{4} = \frac{1}{2}$ (B) $\frac{x}{9} + \frac{y}{4} = 1$ (C) $\frac{9}{x} + \frac{4}{y} = 1$ (D) $\frac{4}{x} + \frac{9}{y} = 1$

[MULTIPLE CORRECT CHOICE TYPE]

Q.15 to Q.17 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.15 Consider the circle $x^2 + y^2 - 10x - 6y + 30 = 0$. Let O be the centre of the circle and tangent at $A(7, 3)$ and $B(5, 1)$ meet at C . Let $S = 0$ represents family of circles passing through A and B , then
 (A) area of quadrilateral $OACB$ equals 4.
 (B) the radical axis for the family of circles $S = 0$ is $x + y = 10$.
 (C) the smallest possible circle of the family $S = 0$ is $x^2 + y^2 - 12x - 4y + 38 = 0$.
 (D) the coordinates of point C are $(7, 1)$.
- Q.16 Let $P(x, y)$ satisfy $x^2 + y^2 + 8x - 10y = 40$.
 If $p = \max\left(\sqrt{(x+2)^2 + (y-3)^2}\right)$ and $q = \min\left(\sqrt{(x+2)^2 + (y-3)^2}\right)$, then
 (A) $p + q = 18$ (B) $p + q = \sqrt{2}$ (C) $p - q = 4\sqrt{2}$ (D) $pq = 73$
- Q.17 The lines $x + y = 1$, $(k - 1)x + (k^2 - 7)y - 5 = 0$ and $(k - 2)x + (2k - 5)y = 0$ are
 (A) concurrent for three values of k (B) concurrent for exactly one value of k
 (C) concurrent for no value of k (D) parallel for $k = 3$

[INTEGER TYPE]

Q.18 & Q.19 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

- Q.18 Let Q and R be the images of a point $P(-3, 1)$ in the mirror $x^2 - y^2 - x + y = 0$. If S is the area of triangle PQR , then find the value of $[\sqrt{S}]$.
 [Note : $[k]$ denotes greatest integer less than or equal to k .]
- Q.19 The line $x + 2y + 3 = 0$ cuts the circle $x^2 + y^2 + 4x + 4y - 1 = 0$ at points P and Q and the line $2x + 3y + \lambda = 0$ cuts the circle $x^2 + y^2 + 6x + 2y - 7 = 0$ at points R and S . If P, Q, R and S are concyclic, then find the value of λ .

ANSWER KEY

Q.1	C	Q.2	A	Q.3	D	Q.4	A	Q.5	D
Q.6	D	Q.7	B	Q.8	A	Q.9	B	Q.10	A
Q.11	C	Q.12	B	Q.13	A	Q.14	C	Q.15	ACD
Q.16	ACD	Q.17	CD	Q.18	[3]	Q.19	[4]		